

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BIOMOLECULES

LECTURE-5

ZOOLOGY

DAVOOD SIR (CEO of NG)

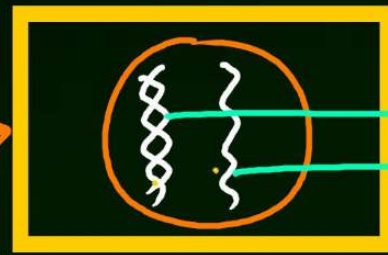


Physics Wallah

Topics To Be Covered

- 1 **Nucleic Acids**
- 2 **Previous Year NEET MCQ's**

Nucleic Acids



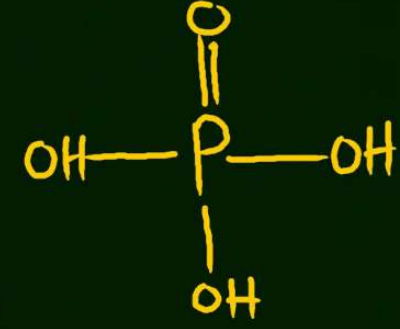
DNA → Deoxyribonucleic acid
RNA → Ribonucleic acid

DNA RNA

Polymers of Nucleotide

COMPOSED

→ Phosphoric acid / phosphate group $[H_3PO_4]$ →

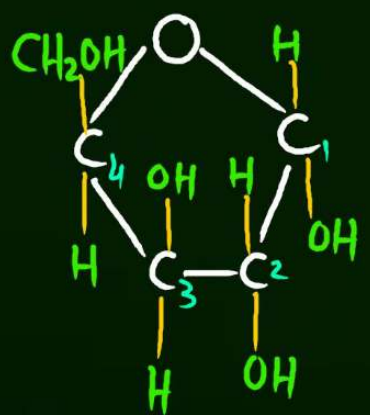


→ Pentose Sugar

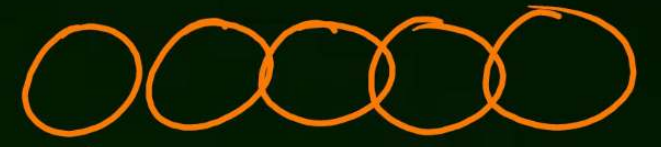
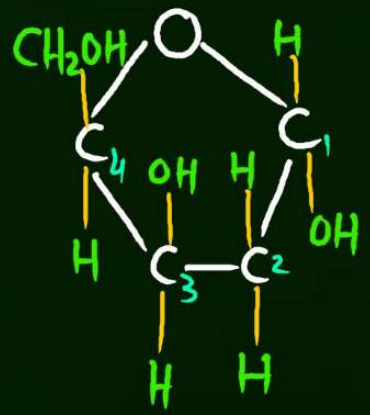
IT IS OF TWO TYPES

→ Nitrogenous base

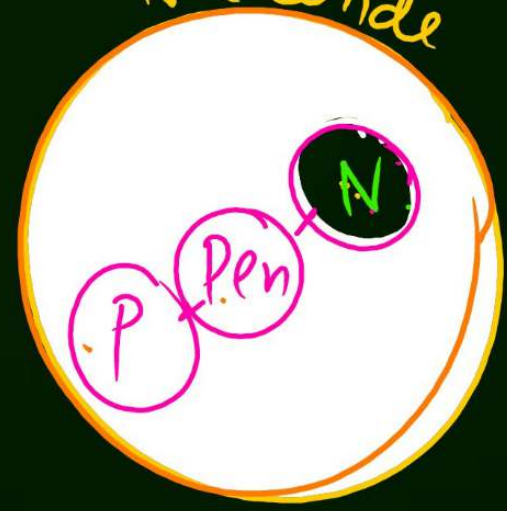
Ribose Sugar
(Present in RNA)



Deoxyribose Sugar
(Present in DNA)



Nucleotide

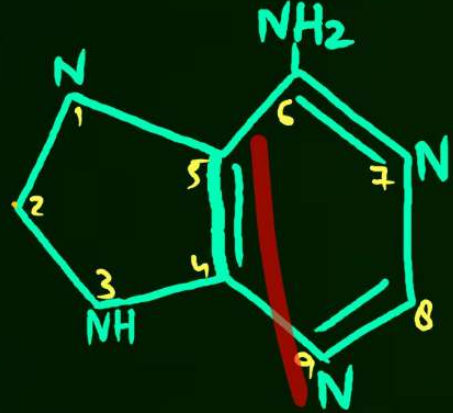


Nucleic Acids

③ Nitrogenous Base

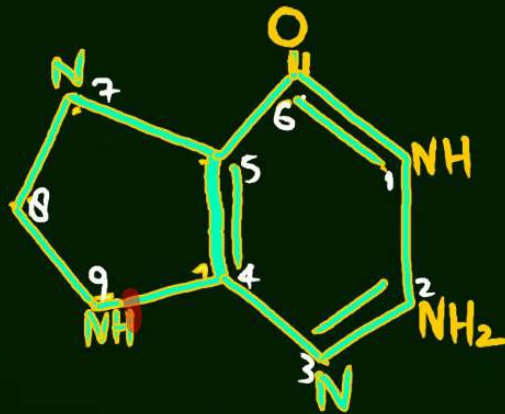
Purines

Adenine



6-Amino-purine

Guanine



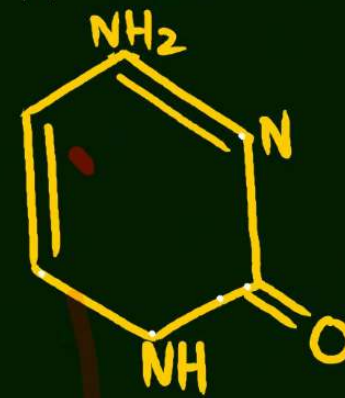
2-amino-6-oxypurine

Pyrimidines

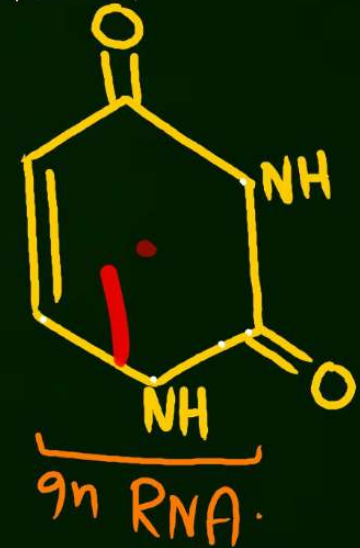
Thymine



Cytosine

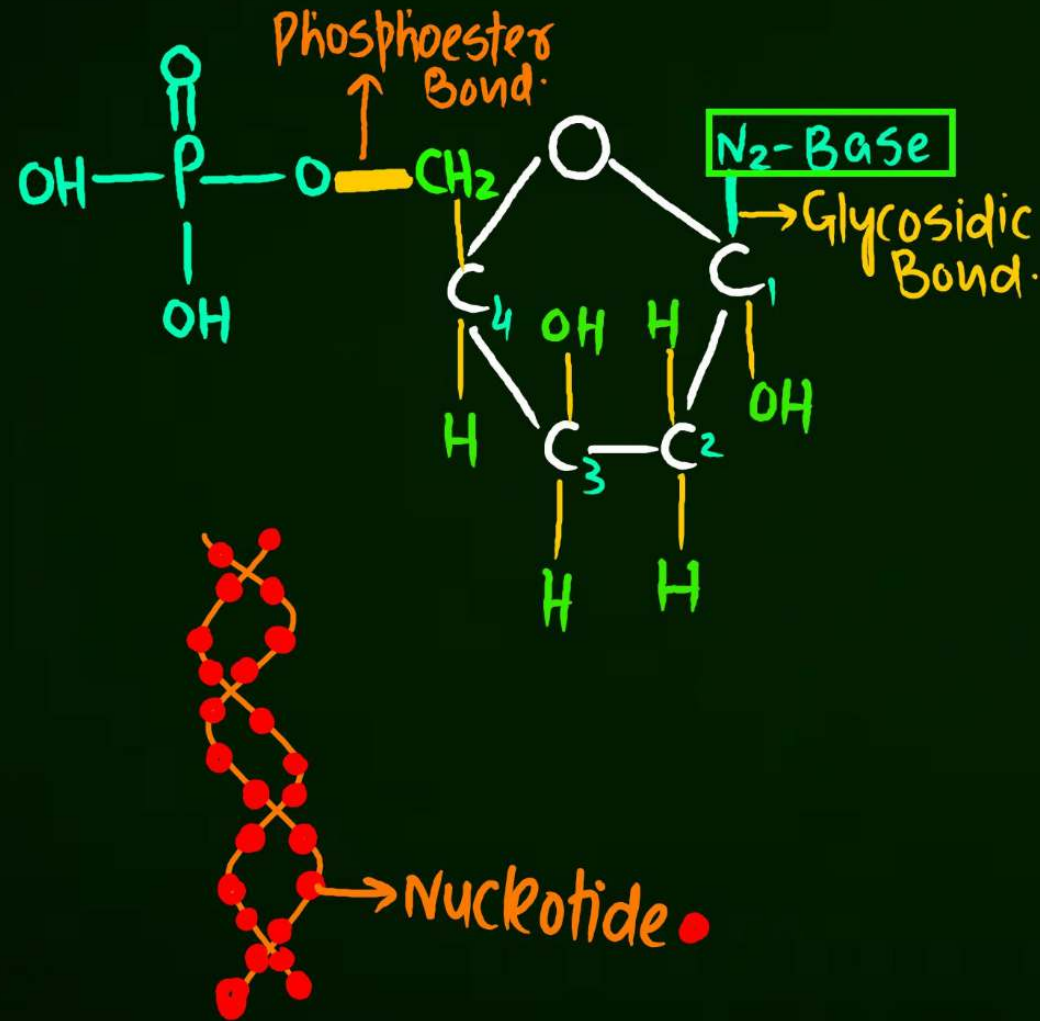


Uracil



Nucleic Acids

Nucleotide



nucleotides are of 5-types depending upon N₂-Base

if Adenine is present	→ Adenylic acid	} Different Nucleotides
if Guanine is present	→ Guanylic acid	
if Thymine is present	→ Thymidylic acid	
if Cytosine is present	→ Cytidylic acid	
if Uracil is present	→ Uridylic acid	

Nucleosides (Pentose sugar + N₂-Base) are 5-types:

Pentose sugar + Adenine	→ Adenosine	} Different Nucleoside
Pentose sugar + Guanine	→ Guanosine	
Pentose sugar + Thymine	→ Thymidine	
Pentose sugar + Cytosine	→ Cytidine	
Pentose sugar + Uracil	→ Uridine (RNA)	

Functions of Nucleic acid

① Heredity

Nucleic acids

Nucleic acids, along with polysaccharides and polypeptides, are macromolecules found in the acid-insoluble fraction of living tissues. Together, they form the **true macromolecular fraction** of any living cell or tissue.

Nucleic acids are **polynucleotides** (long chains of nucleotides), which are their basic building blocks.

Structure of a Nucleotide

Each nucleotide consists of three distinct chemical components:

- 1) *A phosphoric acid or phosphate group*
- 2) *Pentose Sugar (A monosaccharide)*
- 3) *Nitrogenous base (Heterocyclic compound)*

Nucleic acids

1) Nitrogenous Bases:-

- The nitrogen bases in nucleic acids are heterocyclic compounds and are of two types:
 - a) **Purines:-** *Adenine and Guanine.*
 - b) **Pyrimidines:-** *Cytosine, Thymine, and Uracil*

1) Pentose Sugar:-

- The Pentose sugar in the nucleotide can be Ribose sugar (found in RNA) or it can be 2'-Deoxyribose sugar found in DNA . Chemical formula of pentose sugar is $C_5H_{10}O_5$.

3) Phosphoric acid or Phosphate:-

Phosphoric acid (phosphate) in DNA forms the phosphate backbone by linking the 3' carbon of one sugar to the 5' carbon of the next via phosphodiester bonds. It provides structural stability, negatively charges the DNA, and is essential for DNA's double helix structure.

Nucleic acids

➤ Nucleosides and Nucleotides:-

When a phosphate group is also attached (esterified to the sugar), then it becomes a nucleotide, examples:

- Adenylic acid
 - Guanylic acid
 - Cytidylic acid
 - Uridylic acid
 - Thymidylic acid.
- } → Nucleotides.

When a nitrogen base is attached to a Pentose sugar it forms a nucleoside, examples:

- Adenosine
 - Guanosine
 - Cytidine
 - Uridine
 - Thymidine
- } → Nucleosides.

Question [NEET-2026]

Kaun Banayga Nabkaar

Identify the correct statements about biomolecules.

- A. Lipids are generally water soluble. ✗
- B. Proteins are polypeptides. ✓
- C. Polysaccharides are long chains of sugars. ✓
- D. Adenine and Guanine are substituted pyrimidines ✗
- E. Almost all enzymes are proteins. ✓

Choose the correct answer from the options given below.

1 C, D & E only

3 B, D & E only

2 B, C & E only. ✓

4 A, B & C only

Question [NEET-2025]

Kaun Banayga Nabkaar



Match List I with List II.

List-I

A. Adenosine

B. Adenylic Acid

C. Adenine

D. Alanine

List II

I. Nitrogenous base

II. Nucleotide

III. Nucleoside

IV. Amino acid

Choose the correct answer from the options given below.

1 A-III, B-II, C-IV, D-I

2 A-III, B-II, C-I, D-IV

3 A-II, B-III, C-I, D-IV

4 A-III, B-IV, C-II, D-I

Question: NEET-2017

Which of the following is not a polymeric?

- A** Proteins
- B** Polysaccharides
- C** Lipids ✓
- D** Nucleic acids



Thank
You